

How to Select the 'Best' Model?

A Theoretical and Practical Introduction with Information Theory

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March 2025

Course structure

1. Goals and aims of the course
2. The workflow of an ecologist (simplified)
3. Why bother with AIC rather than Null-hypothesis testing?
4. Information Theory: what is the AIC? What does it tell us?
5. Recommendations in practice

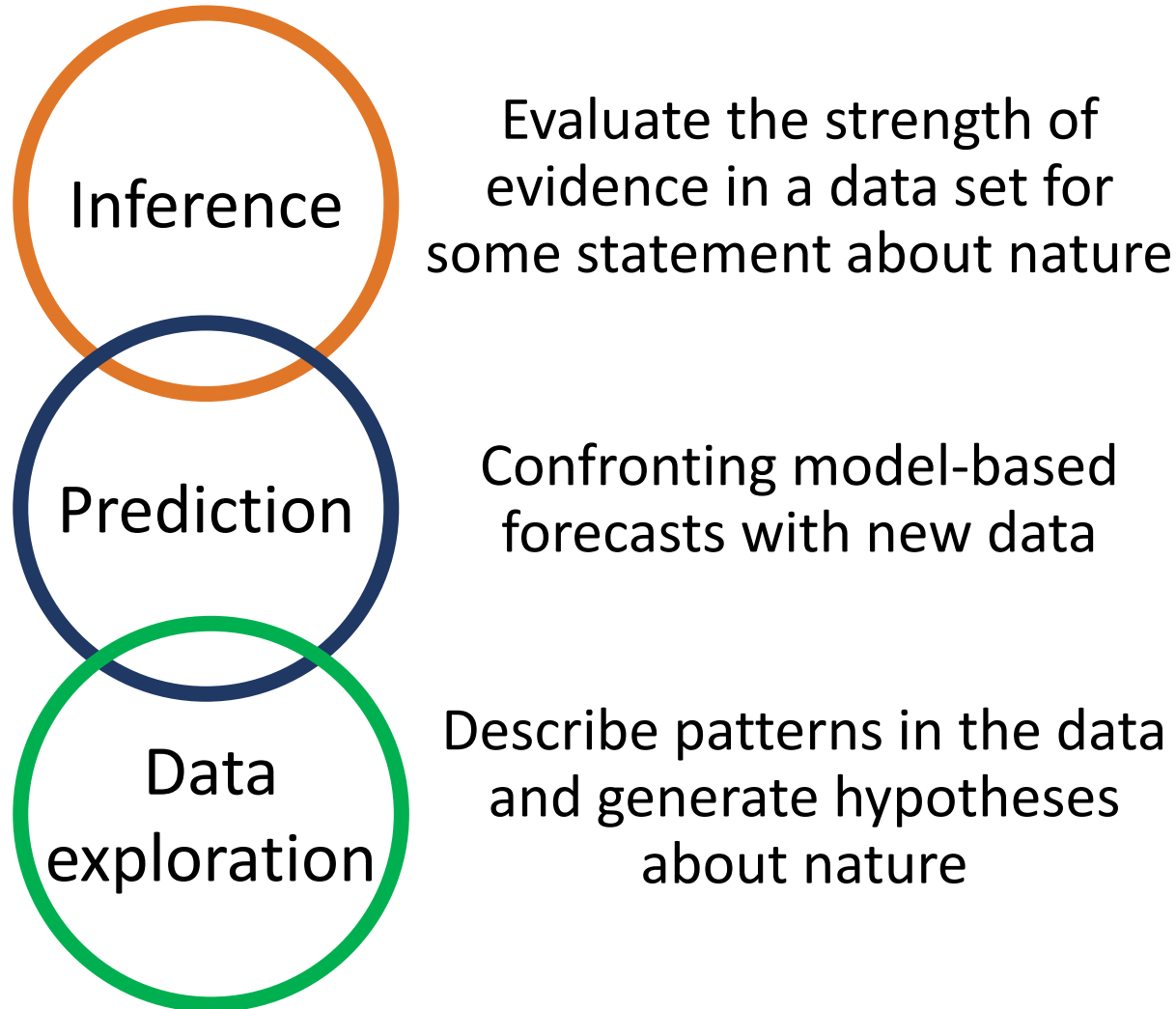
What is this course about?

Statistical methods can be seen as a **toolbox**.
All methods have a theoretical background,
assumptions and limitations.

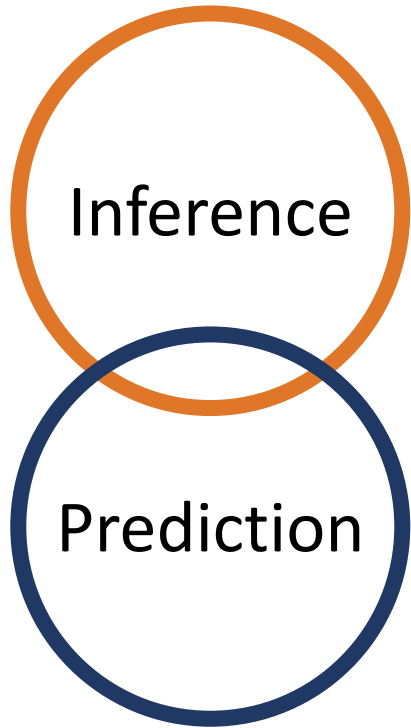
This course aims to help you
answer the question:
**is information theory a good
tool for my research goals?**



What is your goal?



The focus of this course



Evaluate the strength of evidence in a data set for some statement about nature

Confronting model-based forecasts with new data



Which coefficients are non-zero?

Which non-zero effects are positive/negative?

What is the relative importance of variables?



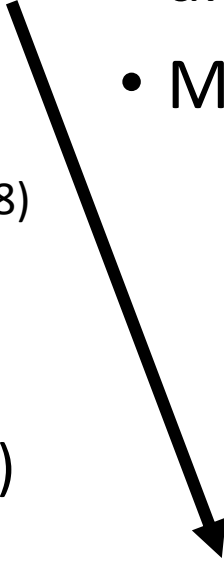
The focus of this course

Inside the scope

- Likelihood-based frequentist methods
 - Linear models (ANOVA, ANCOVA, multiple regression, Kruskal-Wallis; Faraway 2014, 2016)
 - Generalized Linear (Mixed) Models (GL(M)M) (Harrison et al. 2018)
 - Generalized Additive (Mixed) Models (GA(M)M) (Wood 2017)
 - Generalized Least Squares (GLS; Symonds et al. 2014; Fletcher & Fortin 2018)
 - And so on...

Outside the scope

- Bayesian methods (McElreath 2024)
- Causal inference (SEM, path analysis; Pearl & McKenzie 2018; McElreath 2024)
- Machine learning (Pichler & Hartig, 2023)



<https://lindeloev.github.io/tests-as-linear/>

Hypotheses → Models

What set of candidate models
to consider?



Hypotheses → Models

What set of candidate models
to consider?



There is no substitute for
thinking

Include all models that have a
reasonable justification

Based on science, before
examination of the data

Burnham & Anderson 2002, Johnson & Omland 2004



Hypotheses → Models

Richness $\sim$ Rainfall

Richness $\sim$ Rainfall + Disturbance

Richness $\sim$ Soil pH



Hypotheses → Models

Richness $\sim$ Rainfall

Richness $\sim$ Rainfall + Disturbance

Richness $\sim$ Soil pH

Real world → Data



Hypotheses → Models

Richness $\sim$ Rainfall

Richness $\sim$ Rainfall + Disturbance

Richness $\sim$ Soil pH



Real world → Data

- ☞ Model selection
- ☞ Parameter estimation
- ☞ Variable importance
- ☞ Prediction



What approximating model(s) to use for inference?



Occam's razor (Parsimony)

Simpler explanations are preferable.

R^2 would not be appropriate as it does not consider model complexity



How to implement it?

Which tool to use?

Which method to use for
model selection?

Likelihood Ratio Test (LRT)

H0: the 2 models do not differ significantly in log-likelihood (= have similar goodness of fit)

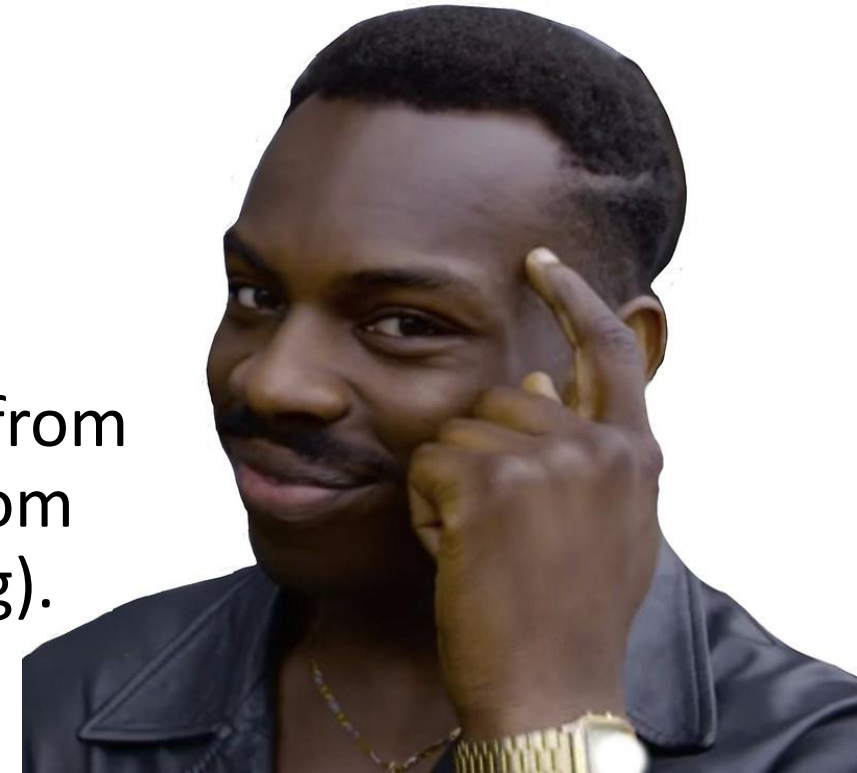
LRT compares pairs of nested models, more complex models are selected if $p < 0.05$, otherwise, we prefer simpler models (parsimony)

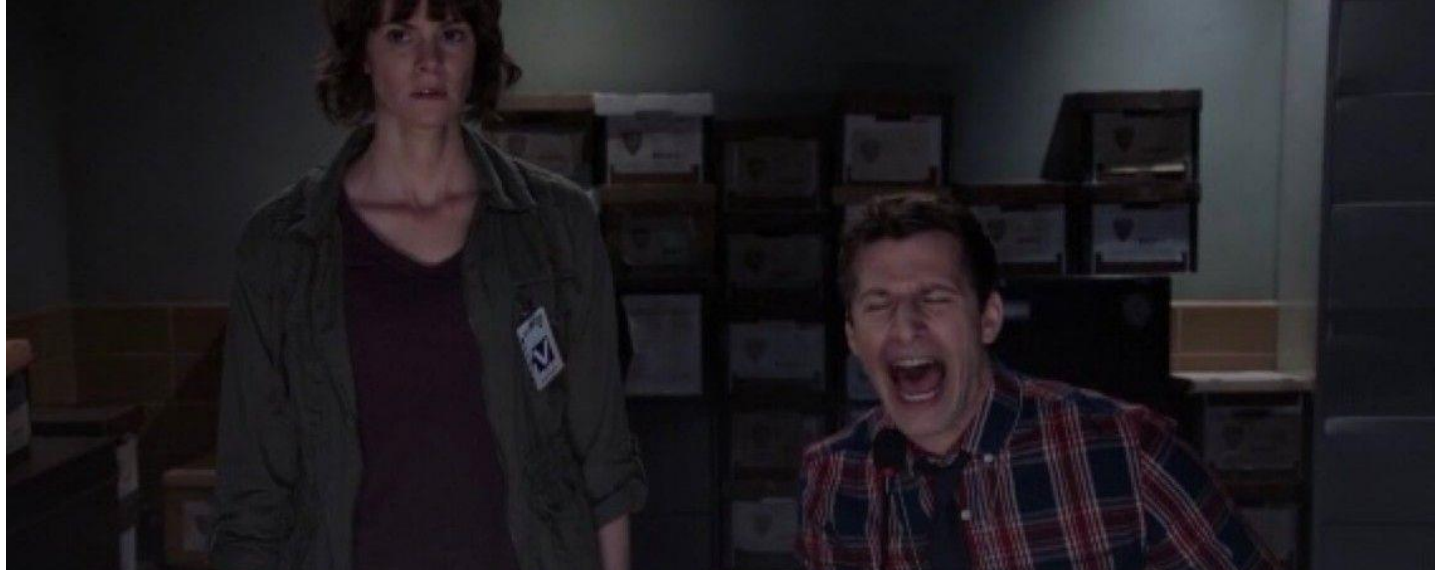
Richness $\sim$ Temperature

Richness $\sim$ Temperature + Soil pH

Is usually implemented as forward (adding variables, from simple to complex), backward (removing variables, from complex to simple), or stepwise (removing and adding).

**Null hypothesis
significance
testing?**





♪ **TELL ME WHY!** ♪



♪ **SHOULD I BOTHER WITH AIC** ♪

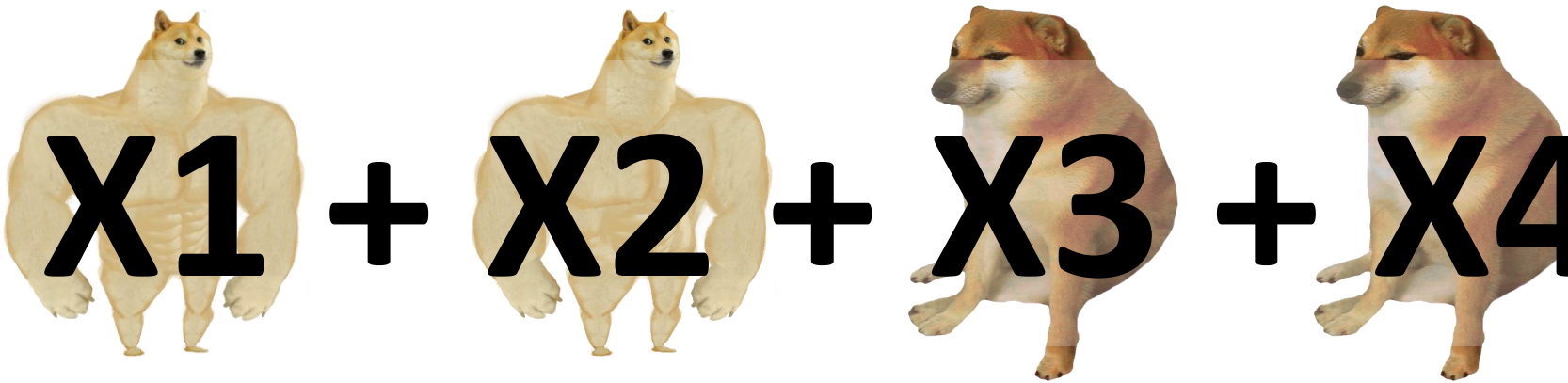
Null-Hypothesis Significance Testing

- ➡ Dichotomic decisions based on the “null-ritual” ($p < 0.05$), rather than “weight of evidence”
- ➡ Only for nested models, cannot quantify relative support among competing models Inference from a single “best model” does not account for model selection bias & uncertainty
- ➡ Results might depend on the order you fit the models
- ➡ Multiple non-independent tests might inflate type I error

Information theory

- ➡ Provides continuous “weight of evidence”, although can also become a ritual ($\Delta AIC < 2$).
- ➡ Quantifies relative support among competing models, nested or non-nested
- ➡ Multimodel inference makes inference on multiple models thus accounting (at least partially) for model selection bias & uncertainty
- ➡ Results do not depend on the order you fit the models
- ➡ No multiple testing

Model selection bias

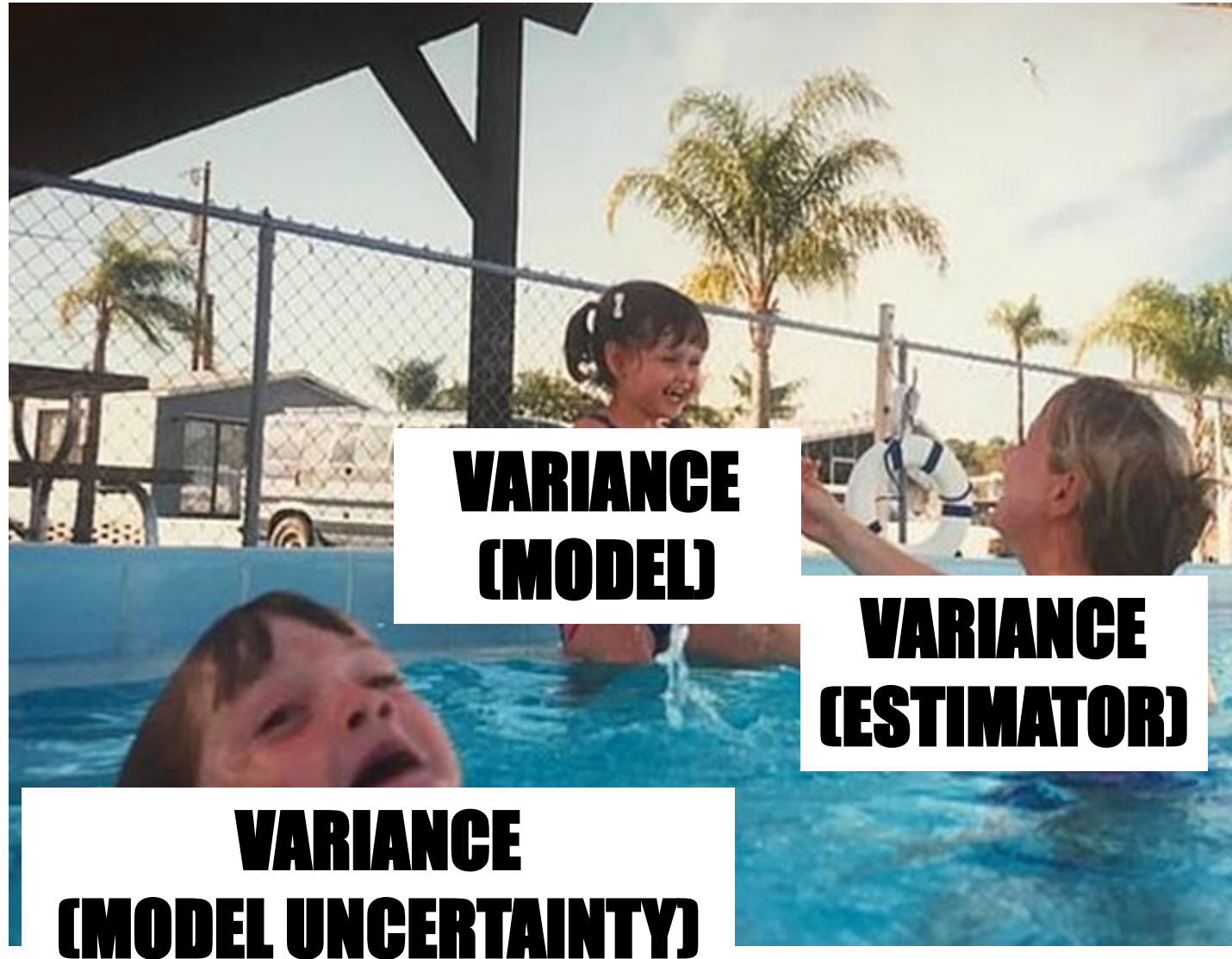
$$Y \sim \text{X1} + \text{X2} + \text{X3} + \text{X4}$$


Selected in all models,
unbiased population inference
 $|\beta| > 0, |\text{estimated } \beta| > 0$

Selected in few models,
biased population inference
 $|\beta| \approx 0, |\text{estimated } \beta| > 0$

- ➡ **Tapering of effect sizes:** effect sizes of decreasing importance are gradually revealed as sample size increases
- ➡ **Collinearity** can also bias coefficients

Model selection uncertainty



“Model selection uncertainty is the component of variance that reflects that model selection merely estimates which model is best, based on the single data set; a different model (in the fixed set of models considered) may be selected as best for a different replicate data set arising from the same experiment.”

Null Hypothesis Significance
testing has some value
(Not everyone agrees! What a surprise!)



In terms of model selection
though, it looks like a poor
alternative to Information
Criteria

Burnham & Anderson 2002, Stephens
et al. 2005, 2007, Lukacs et al 2007

NULL HYPOTHESIS TESTING IS NOT ENOUGH!



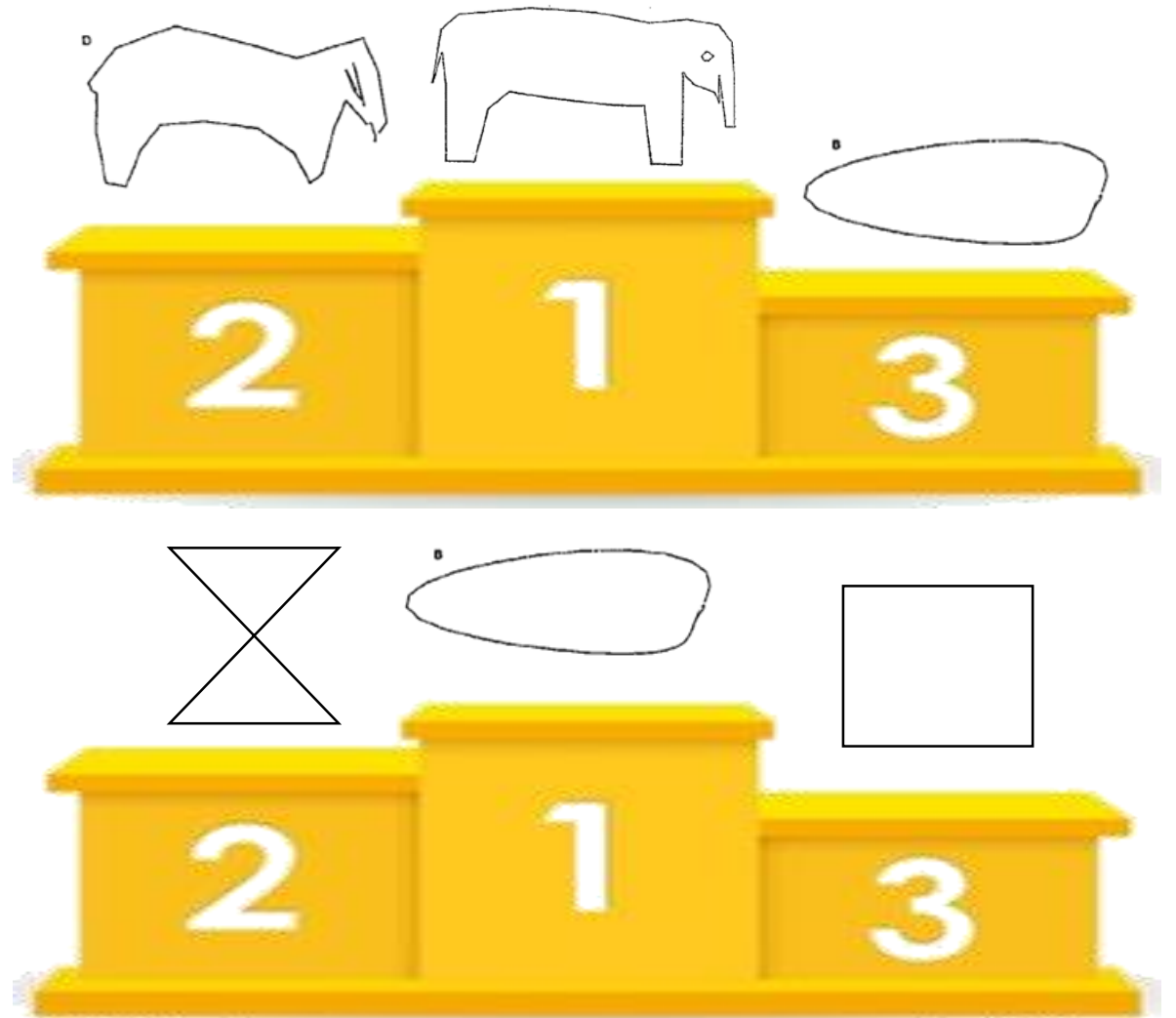
CASE CLOSED!!

Model selection with Information Theory (IT)

Each model is an approximation of truth
Information Theory (IT) allows to rank models
using information criteria



IT will always find a “best” model,
but maybe the “best” model is rubbish

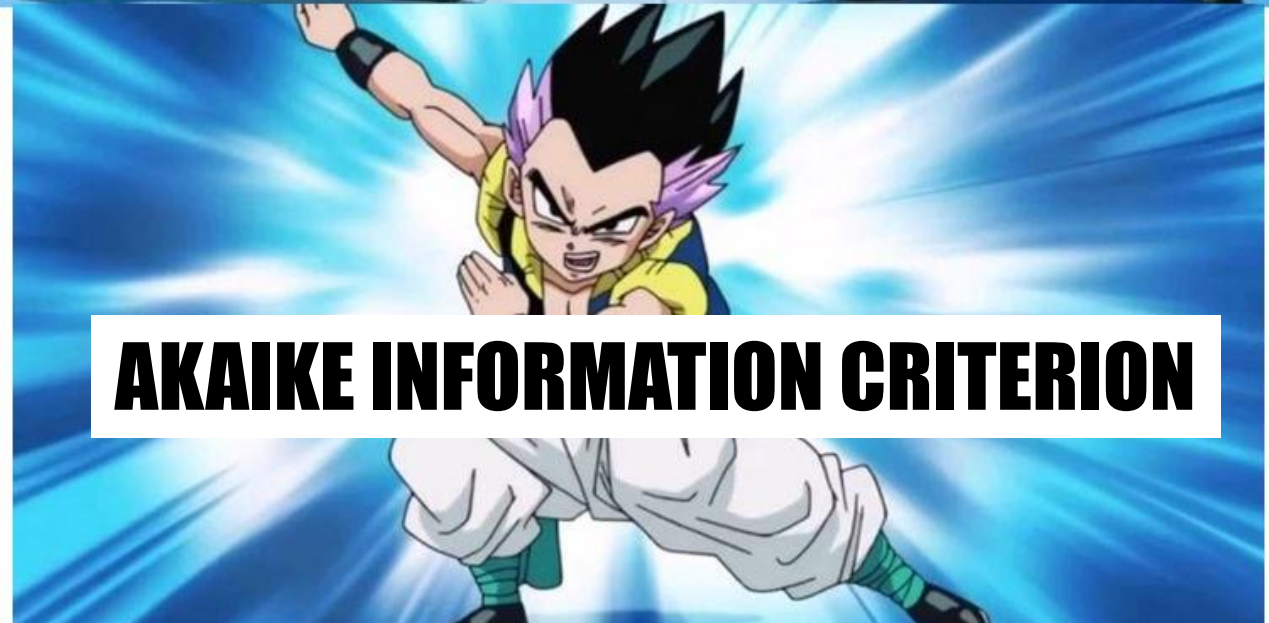


Where does AIC come from?
What does it tell us?

Burnham &
Anderson 2002



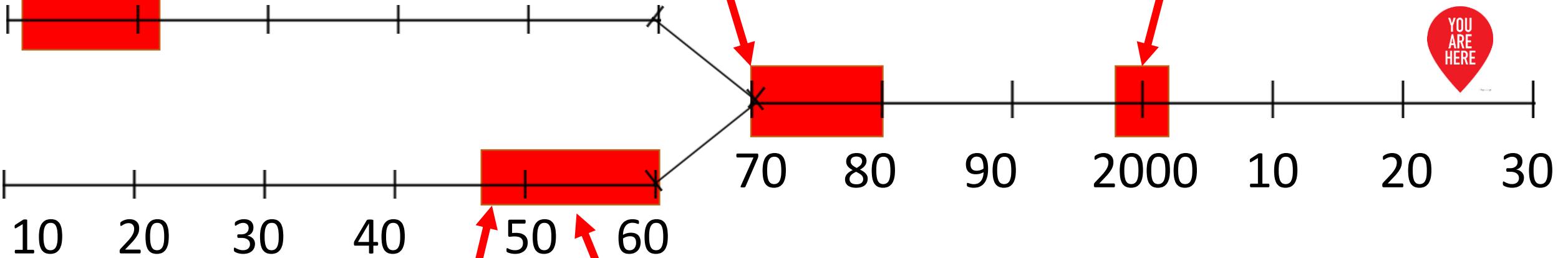
Hirotugu Akaike
(1927-2009)



Maximum likelihood
(Ronald Fisher)

Akaike Information Criterion
(Hirotugu Akaike)

IT methods gain traction
(Kenneth Burnham, David Anderson)



Information theory
(Claude Shannon)

Kullback-Leibler distance
(Solomon Kullback, Richard Leibler)

Burnham &
Anderson 2002

Kullback-Leibler distance

- ➡ Discrepancy, oriented or directed distance (not a simple distance)
- ➡ $I(f, g)$: loss of information when g (model) approximates f (truth)
- ➡ Always positive (or 0 when $g = f$)

$$I(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x|\theta)} \right) dx$$



Maximum log-likelihood

$$L(\text{parameters} \mid \text{data, model})$$

Likelihood(unknown $\mid$ given, assumed)

- ➡ Likelihood is a function only of the unknown parameters
- ➡ Likelihood is a **relative value**

Maximum log-likelihood



Coin toss,
7 successes out of
11 trials
binomial model
 $p = 0.63$

$$g(y, n | p, \text{binomial}) = \binom{n}{y} p^y (1 - p)^{n-y}$$

$$\mathcal{L}(p | y, n, \text{binomial}) = \binom{n}{y} p^y (1 - p)^{n-y}$$

$$\begin{aligned} \log(\mathcal{L}(p | y, n, \text{binomial})) &= \log \binom{n}{y} + y \cdot \log(p) + (n - y) \cdot \log(1 - p), \\ &= \log \binom{11}{7} + 7 \cdot \log(p) + (11 - 7) \cdot \log(1 - p) \\ &= 5.79909 + 7 \cdot \log(p) + (4) \cdot \log(1 - p). \end{aligned}$$

Maximum log-likelihood

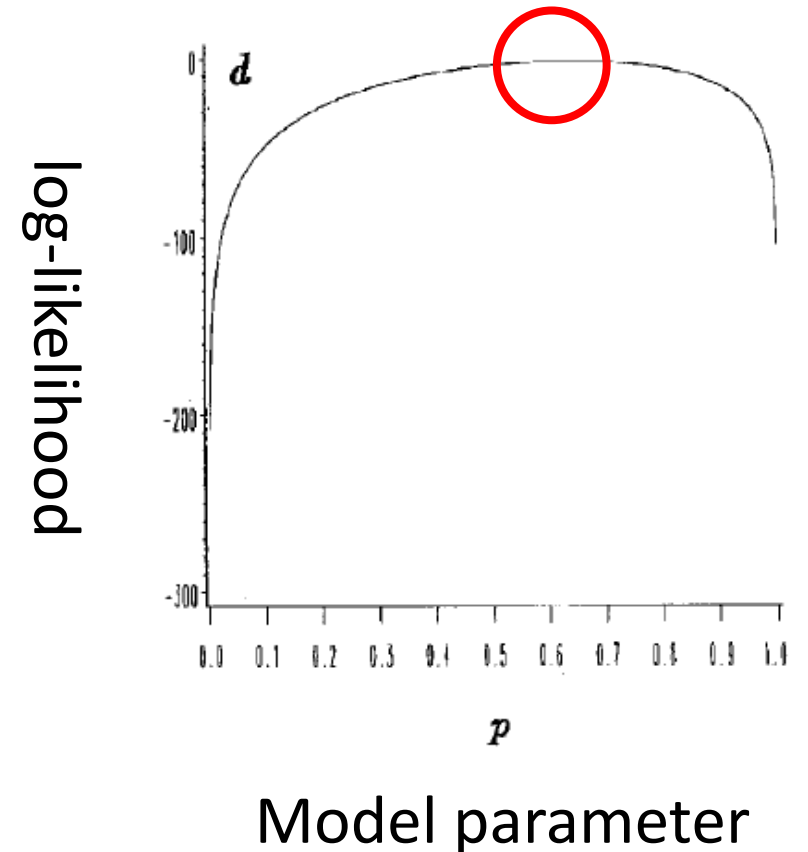


Coin toss,
7 successes out of
11 trials
binomial model
 $p = 0.63$

Maximum likelihood

Find the parameter
value that maximizes
the log-likelihood
function*

- * for a given data
and model
- * iterative methods



f = truth
g = model
I = information lost



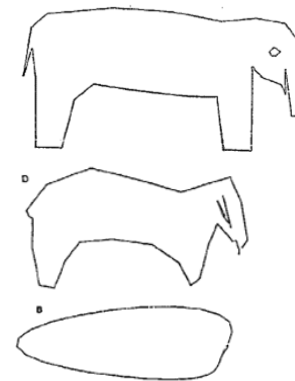
$$I(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x|\theta)} \right) dx \longleftarrow \text{Kullback-Leibler distance}$$

$$I(f, g) = \int f(x) \log(f(x)) dx - \int f(x) \log(g(x | \theta)) dx.$$

$$I(f, g) = E_f [\log(f(x))] - E_f [\log(g(x | \theta))]$$

$$I(f, g) = C - E_f [\log(g(x | \theta))],$$

$$I(f, g) - C = -E_f [\log(g(x | \theta))].$$



3 competing
models
(g(x|θ))

f = truth
g = model
I = information lost

$$I(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x|\theta)} \right) dx$$

Kullback-Leibler
distance

Number of parameters as an
asymptotic bias correction

$$I(f, g) - C = -E_f[\log(g(x | \theta))].$$

$$- \hat{E}_{\hat{\theta}}[I(f, \hat{g})] + \mathbf{C} = \log(\mathcal{L}(\hat{\theta} | data)) - K$$

f = truth
g = model
I = information lost

$$I(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x|\theta)} \right) dx \longleftarrow \text{Kullback-Leibler distance}$$

Burnham & Anderson (2002) p.64: “As long as *both* terms (the log-likelihood and the bias correction) are multiplied by the same negative constant, the model where the criterion is minimized is unchanged and there is nothing arbitrary.”

$$I(f, g) - C = -E_f[\log(g(x | \theta))].$$

$$- \hat{E}_{\hat{\theta}}[I(f, \hat{g})] + C = \log(\mathcal{L}(\hat{\theta}|data)) - K$$

$$-2 \log(\mathcal{L}(\hat{\theta}|y)) + 2K = \text{AIC}$$

x -2

Estimate of the expected, **relative** distance between the fitted model and the unknown true mechanism that generated the observed data

f = truth
 g = model
 I = information lost

$$I(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x|\theta)} \right) dx$$

Kullback-Leibler distance

Unknown, Variable Unknown, Constant Known, Variable

$$-\hat{E}_{\hat{\theta}}[I(f, \hat{g})] + C = \log(\mathcal{L}(\hat{\theta}|data)) - K$$

$$-2 \log(\mathcal{L}(\hat{\theta}|y)) + 2K = \text{AIC}$$

x -2

f = truth
g = model
I = information lost

$$I(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x|\theta)} \right) dx$$

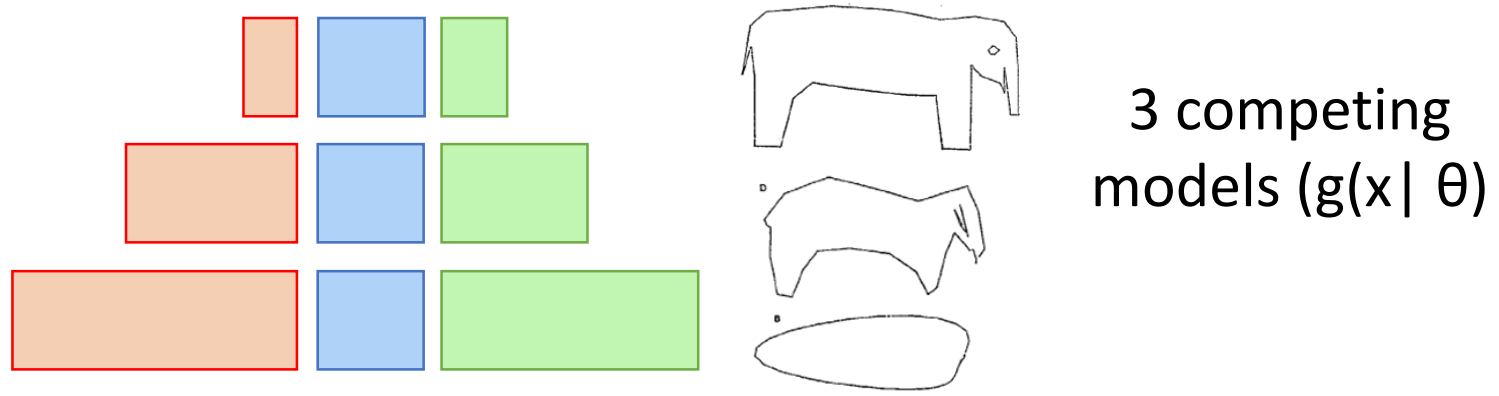
Kullback-Leibler distance

Unknown, Variable Unknown, Constant Known, Variable

$$-\hat{E}_{\hat{\theta}}[I(f, \hat{g})] + \mathbf{C} = \log(\mathcal{L}(\hat{\theta}|data)) - K$$

$$-2 \log(\mathcal{L}(\hat{\theta}|y)) + 2K = \mathbf{AIC}$$

x -2





ONLY RELATIVE DIFFERENCES MATTER.

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DIFFERENCES MATTER.

ONLY RELATIVE

AIC Differences

It is not the absolute size of the AIC value, it is the relative values, and particularly the AIC differences (Δ_i), that are important.

An individual AIC value, by itself, is not interpretable due to the unknown constant (interval scale). AIC is only comparative, relative to other AIC values in the model set; thus such differences Δ_i are very important and useful.

Burnham&Anderson (2002) p.71

ONLY RELATIVE DIFFERENCES MATTER.

f = truth
 g = model
 I = information lost

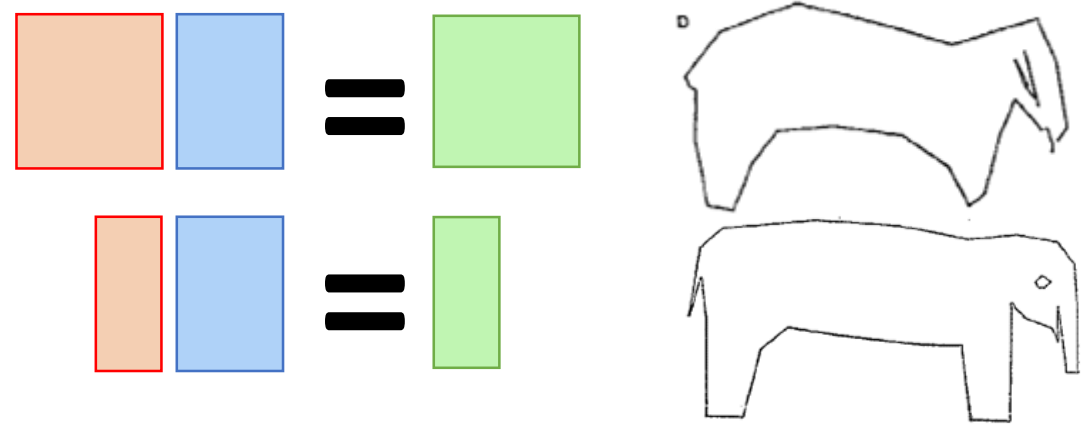
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Kullback-Leibler distance

$$-\hat{E}_{\hat{\theta}}[I(f, \hat{g})] + C = -2 \log(\mathcal{L}(\hat{\theta}|y)) + 2K = \text{AIC}$$

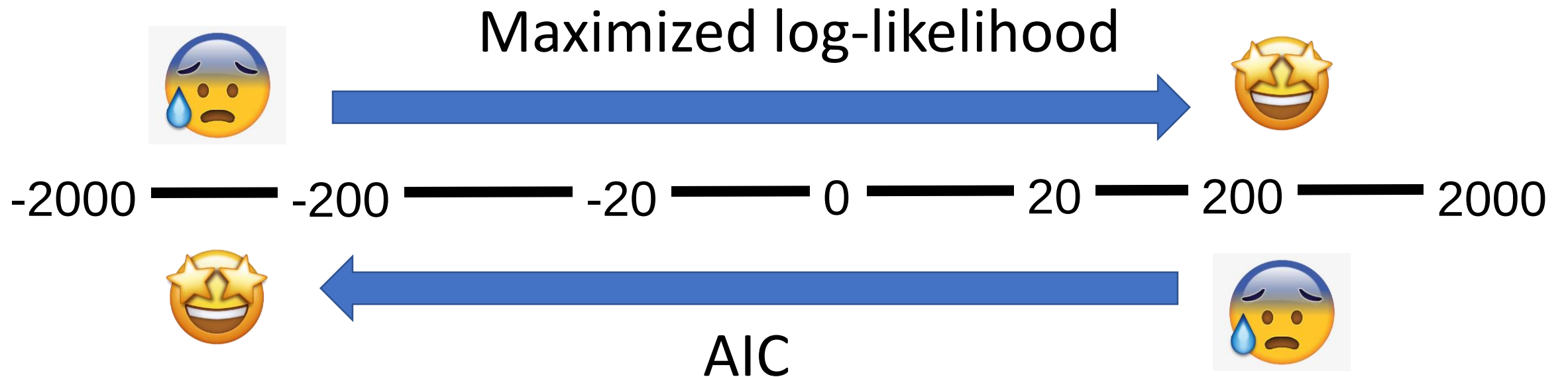
Unknown, Variable Unknown, Constant Known, Variable

When you calculate ΔAIC , "truth" disappears as a constant



AIC1
 AIC2

- 👉 Information Criteria means the same anywhere on the interval (relative) scale
- 👉 No thresholds! 0 means nothing, negative values mean nothing



$$\text{AIC} = -2 \log(\mathcal{L}(\hat{\theta}|y)) + 2K$$



$$-196 = -2 * 100 + 2 * 2$$

$$-180 = -2 * 100 + 2 * 10$$



$$204 = -2 * (-100) + 2 * 2$$

$$220 = -2 * (-100) + 2 * 10$$



**“WITHIN A SET OF COMPETING MODELS,
THE SMALLER,
THE BETTER”**



**“WITHIN A SET OF COMPETING MODELS,
THE CLOSER TO $-\infty$,
THE BETTER”**

Important points about the AIC

- ☞ AIC is only an **estimate** of the relative Estimated Kullback-Leibler distance (EKLD); because of **sampling error**, a model not having the lowest AIC value may in fact be the best EKLD model
- ☞ Thus, **Information Criteria (IC) are highly dataset-specific**, and therefore not comparable across different datasets, nor useful for meta-analysis
- ☞ On increasing sample size, AIC will tend to select more complex models, specially if they incorporate some important process -> with greater sample size, the tapering effect sizes are revealed

Important points about the AIC

- 👉 The **data and response variables must be fixed**. For instance, do not compare: response VS $\log(\text{response})$, dataset with outliers VS dataset without outliers
- 👉 Information Criteria have been discussed in the context of selecting the model's distribution (e.g. Poisson vs Negative Binomial, Blasco-Moreno et al. 2019; Campbell 2021).
- 👉 **Information Theory (IT) is not a test**. It is advised not to perform model selection through Null Hypothesis Significance Testing to simplify a parsimonious model obtained through IT.

OK, so ΔAIC can help rank models from best to worst

What next?

Once ranked according to ΔAIC , a threshold can be chosen to select which model(s) to draw inference from

AIC	ΔAIC
- 2	$-2 - (-2) = 0$
- 1	$-1 - (-2) = 1$
0	$0 - (-2) = 2$

Usually, a threshold of 2 is used to select models that are “equally supported” from an Information Theory perspective. Some simulations suggest that a threshold of at least six is required so that the probability of keeping the “best” Kullback-Leibler model becomes 95% (it can be ~80% for a threshold of 2)

Once ranked according to ΔAIC , a threshold can be chosen to select which model(s) to draw inference from

ΔAIC : 0-2

Substantial
empirical support



ΔAIC : 4-7

Considerably less
empirical support



$\Delta AIC > 10$

Essentially no
empirical support



Likelihood of a model, Given data

$$\mathcal{L}(g_i|x) \propto \exp\left(-\frac{1}{2}\Delta_i\right)$$

Relative strength of evidence for each model:
a quantification of the plausibility of each
model as being the actual K-L best model (not
normalized, does not sum 1)

AIC	Δ AIC	L(model data)
- 2	$-2 - (-2) = 0$	$e^{(-0.5*0)} = 1$
- 1	$-1 - (-2) = 1$	$e^{(-0.5*1)} = 0.61$
0	$0 - (-2) = 2$	$e^{(-0.5*2)} = 0.37$

Akaike weights

$$w_i = \frac{\exp(-\frac{1}{2}\Delta_i)}{\sum_{r=1}^R \exp(-\frac{1}{2}\Delta_r)}$$

Probability (weight of evidence) that a model is the actual Kullback-Leibler best model, within the set of candidate models. Basically, we normalize relative likelihoods to sum up to 1.

AIC	Δ AIC	L(model data)	Akaike weight (w)
- 2	$-2 - (-2) = 0$	$e^{(-0.5*0)} = 1$	$1 / 1.98 = 0.5$
- 1	$-1 - (-2) = 1$	$e^{(-0.5*1)} = 0.61$	$0.61 / 1.98 = 0.31$
0	$0 - (-2) = 2$	$e^{(-0.5*2)} = 0.37$	$0.37 / 1.98 = 0.19$

Akaike weights

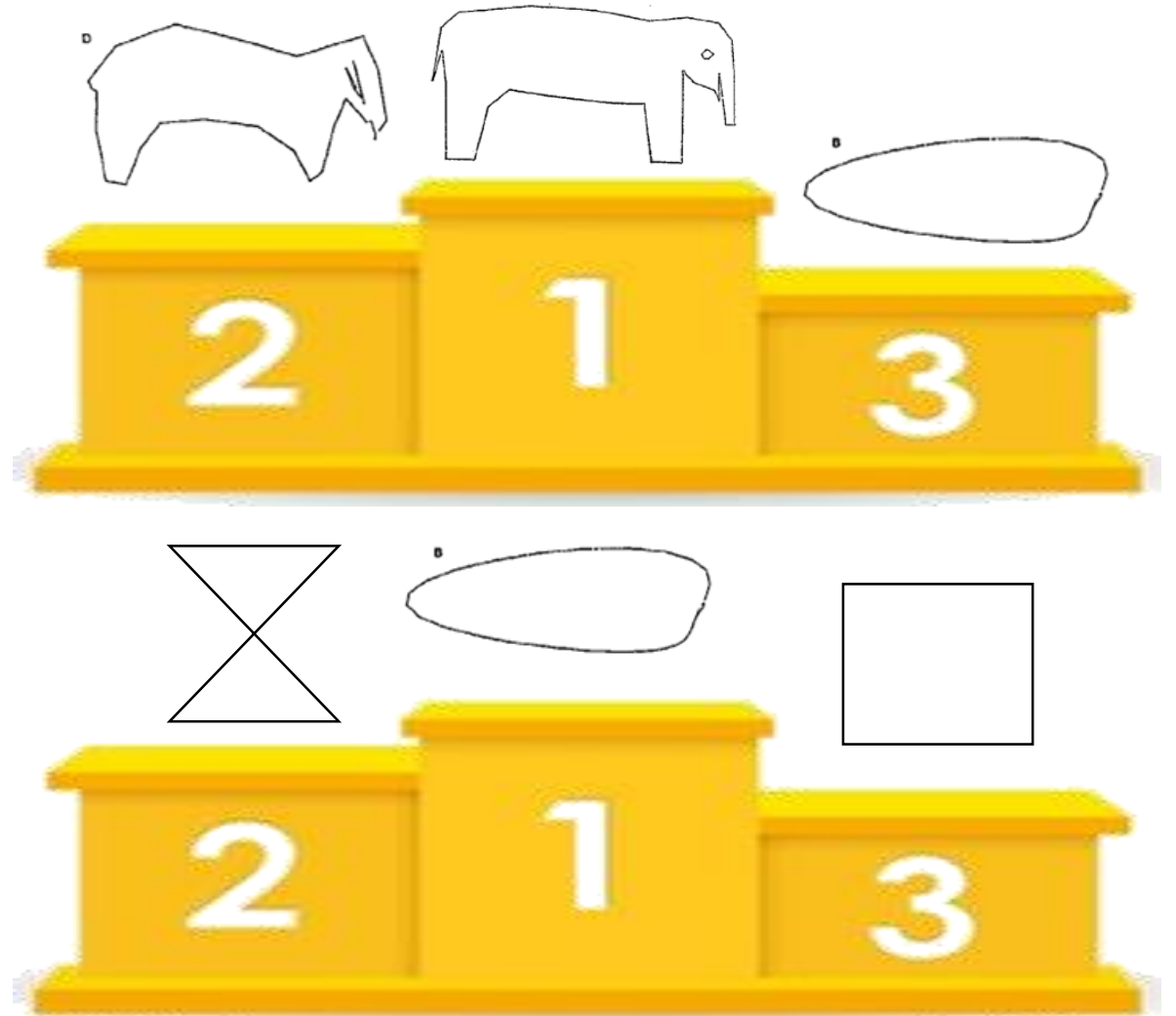
$$w_i = \frac{\exp(-\frac{1}{2}\Delta_i)}{\sum_{r=1}^R \exp(-\frac{1}{2}\Delta_r)}$$

Probability (weight of evidence) that a model is the actual Kullback-Leibler best model, within the set of candidate models. Basically, we normalize relative likelihoods to sum up to 1.

AIC	Δ AIC	L(model data)	Akaike weight (w)		
- 2	$-2 - (-2) = 0$	$e^{(-0.5*0)} = 1$	$1 / 1.98 = 0.5$	X1	X1 = 0.81
- 1	$-1 - (-2) = 1$	$e^{(-0.5*1)} = 0.61$	$0.61 / 1.98 = 0.31$	X1 + X2	X2 = 0.31
0	$0 - (-2) = 2$	$e^{(-0.5*2)} = 0.37$	$0.37 / 1.98 = 0.19$	X3	X3 = 0.19

The probability that a given predictor variable appears in the best model is estimated by summing the weights of each model where the variable appears. This has been used as a metric of relative importance.

IT will always find a “best” model,
but maybe the “best” model is rubbish



Evidence ratios

$$\mathcal{L}(g_i|x)/\mathcal{L}(g_j|x)$$

Evidence about fitted models about which is better in a Kullback-Leibler information sense. They do not depend on the full set of models, just on the two models being compared.

AIC	Δ AIC	L(model data)	Akaike weight (w)	Evidence ratio (wi/wj)
- 2	$-2 - (-2) = 0$	$e^{(-0.5*0)} = 1$	$1 / 1.98 = 0.5$	
- 1	$-1 - (-2) = 1$	$e^{(-0.5*1)} = 0.61$	$0.61 / 1.98 = 0.31$	$0.5 / 0.31 = 1.61$
0	$0 - (-2) = 2$	$e^{(-0.5*2)} = 0.37$	$0.37 / 1.98 = 0.19$	$0.5 / 0.19 = 2.7$

Evidence ratios

$$\mathcal{L}(g_i|x)/\mathcal{L}(g_j|x)$$

Evidence about fitted models about which is better in a Kullback-Leibler information sense. They do not depend on the full set of models, just on the two models being compared.

Δ_i	Level of Empirical Support	Δ_j	Evidence ratio
0–2	Substantial	2	2.7
4–7	Considerably less	4	7.4
> 10	Essentially none	8	54.6
		10	148.4
		15	1,808.0
		20	22,026.5

If the best-ranked model has a weight < 1 , it suggests uncertainty about which is the “best” model, and therefore interpretation could rely on a set of competing models

AIC	ΔAIC	$L(\text{model} \text{data})$	Akaike weight (w)	Evidence ratio (w_i/w_j)
- 2	$-2 - (-2) = 0$	$e^{(-0.5*0)} = 1$	$1 / 1.98 = 0.5$	
- 1	$-1 - (-2) = 1$	$e^{(-0.5*1)} = 0.61$	$0.61 / 1.98 = 0.31$	$0.5 / 0.31 = 1.61$
0	$0 - (-2) = 2$	$e^{(-0.5*2)} = 0.37$	$0.37 / 1.98 = 0.19$	$0.5 / 0.19 = 2.7$

Multimodel inference allows to make inference from a set of competing models

Natural averaging (=Unconditional)

- ➡ Models without the variable are used in the averaging
- ➡ The coefficient may shrink towards 0

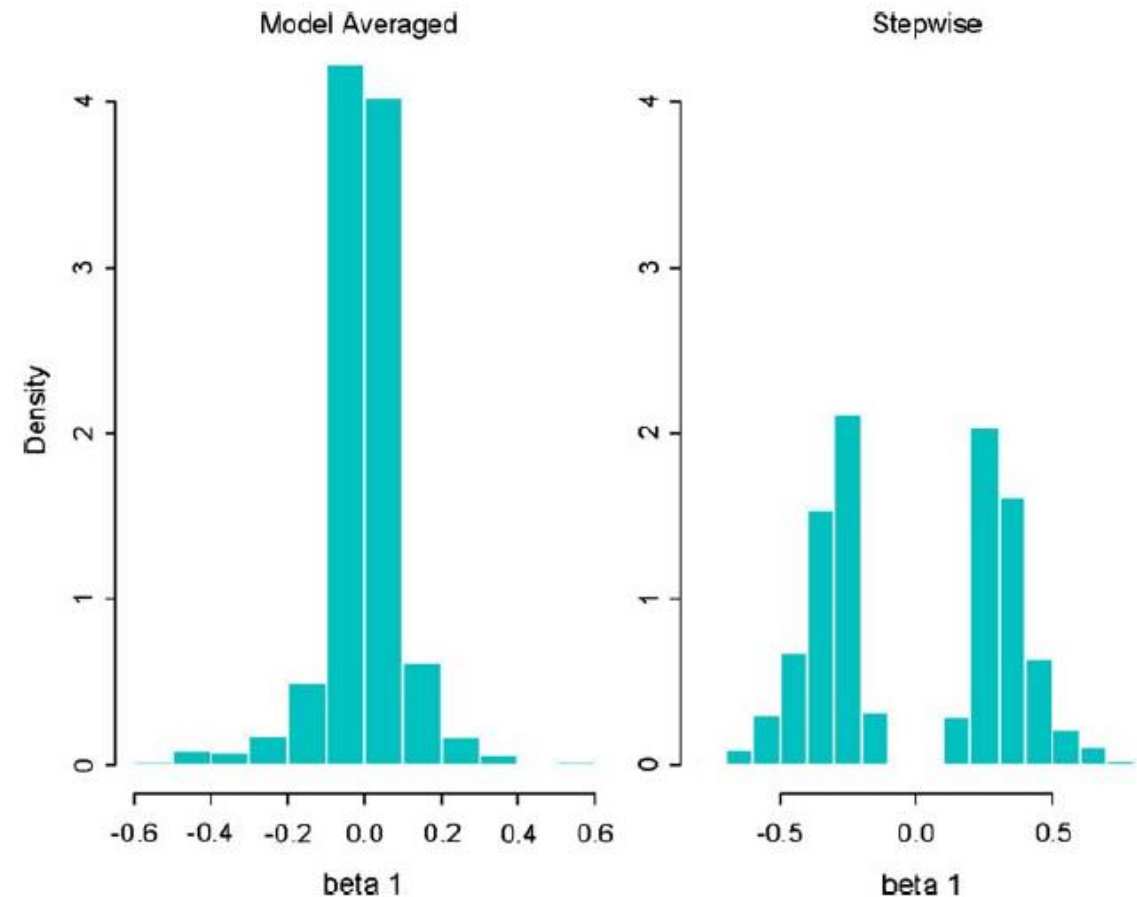
Conditional averaging

- ➡ Models without the variable are not used in the averaging
- ➡ No shrinking towards 0

Model-averaged coefficients and model-averaged standard errors are obtained, using the Akaike weights

A simulation example of unconditional averaging vs stepwise model selection

- ➡ 1000-10000 simulations, $N = 40$, 20 uncorrelated explanatory variables
- ➡ Model-averaged estimates were better than those from the best stepwise models, which selected spurious variables
- ➡ Confidence interval coverage for the model-averaged parameters were close to the nominal 95% level, while stepwise were only 41.9%



Recommendations in practice

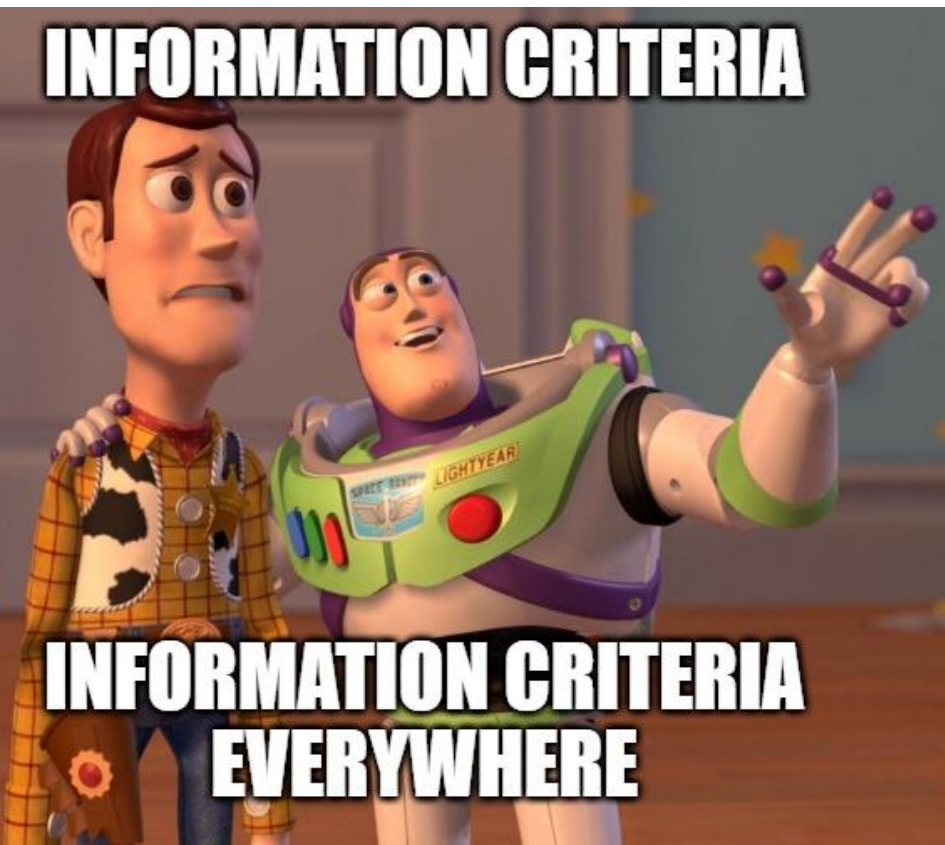
About the importance of
thinking

It is the scientist's responsibility
to include ecologically meaningful models
in the **candidate model set**

- 👉 **Our candidate set of models might not include the “true one”, nor all the parameters that generated the data**



About the importance of



choosing an
appropriate
information criterion

Other Information Criteria than AIC

👉 **Akaike information criterion corrected for small sample size = second order = AICc**

👉 Adequate for small sample sizes, e.g. $n/k < 40$

$$\text{AIC}_c = -2 \log(\mathcal{L}(\hat{\theta})) + 2K \left(\frac{n}{n - K - 1} \right)$$

$$\text{AIC}_c = \text{AIC} + \frac{2K(K + 1)}{n - K - 1}$$

Other Information Criteria than AIC

👉 QuasiAIC(c)

- 👉 Overdispersed responses modelled with quasi-Poisson/Binomial (Richards 2008).
- 👉 K includes overdispersion parameter (c)
- 👉 Consider that alternatives exist to the Poisson (negative binomial, Conway-Maxwell-Poisson, Ver Hoef & Boveng 2007, Faraway, 2016) and Binomial (beta-binomial, Faraway 2016)
- 👉 Recent papers discuss how to select between Poisson models and Negative Binomial models (Blasco-Moreno et al. 2019; Campbell 2021)

$$\text{QAIC} = -[2 \log(\mathcal{L}(\hat{\theta}))/\hat{c}] + 2K,$$

$$\begin{aligned}\text{QAIC}_c &= -[2 \log(\mathcal{L}(\hat{\theta}))/\hat{c}] + 2K + \frac{2K(K+1)}{n-K-1}, \\ &= \text{QAIC} + \frac{2K(K+1)}{n-K-1}\end{aligned}$$

Other Information Criteria than AIC

👉 **Bayesian information criterion = Schwartz's information criterion = BIC**

- 👉 Not based on Kullback-Leibler distance
- 👉 Assumes truth is in the set of candidate models
- 👉 Tends to select simpler models than AIC, particularly as sample size increases
- 👉 Is consistent, unlike AIC, which means that, as N approaches infinity, the probability of BIC of selecting a true low dimensional model approaches 1 (more on this later)

$$-2 \log(\mathcal{L}) + K \cdot \log(n)$$

AIC vs BIC

Aho et al. 2014

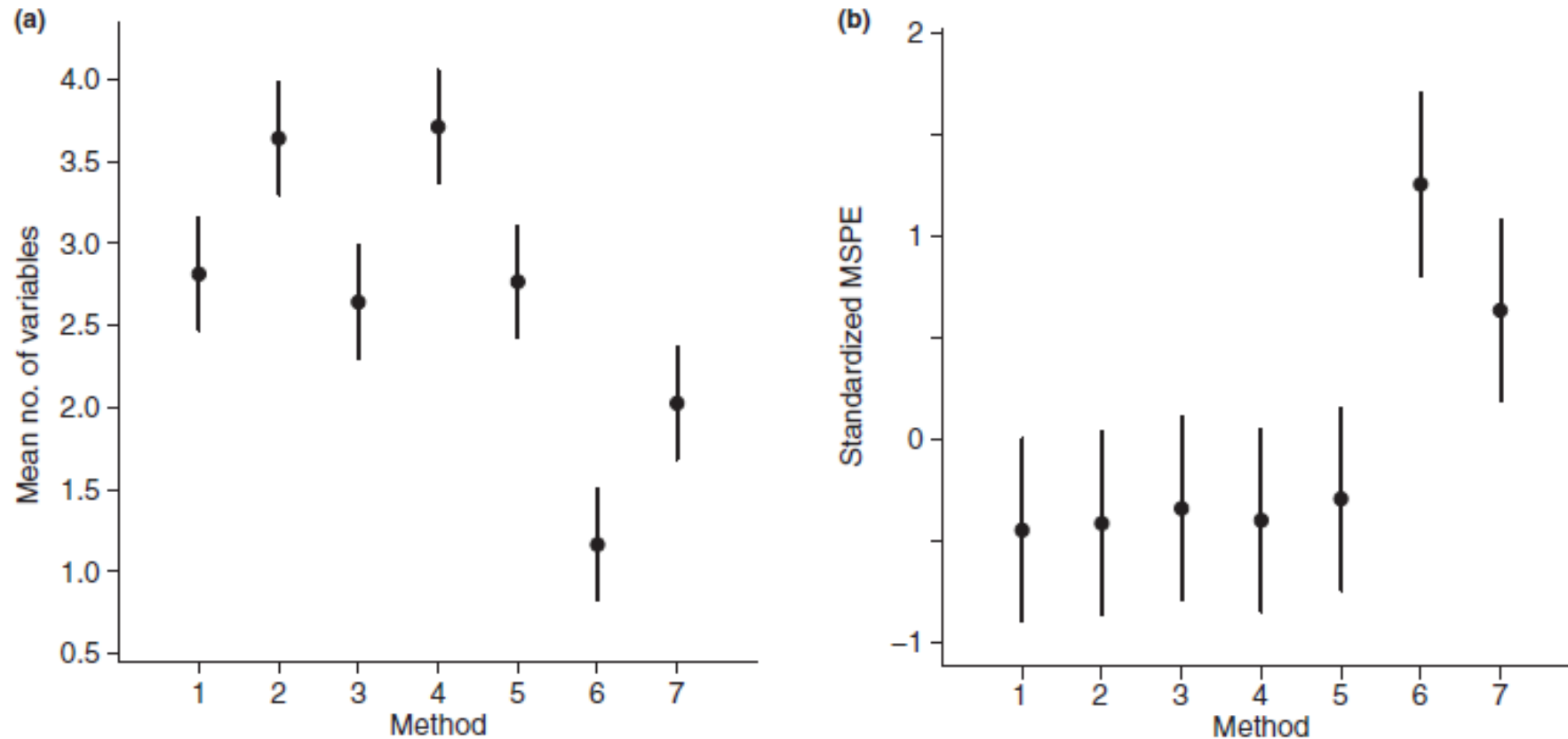
TABLE 2. The worlds of AIC and BIC contrasted.

Factor	AIC	BIC
Mathematical characteristics		
Derivation	Estimated information loss.	Approximate Bayes factor.
Optimality criterion	Asymptotic efficiency.	Consistency.
Close cousins	Data splitting, Mallows' C_p , PRESS.	Hannan-Quinn, Geweke and Meese, Bayes factors, and Bayesian hypothesis testing.
World View		
Problem statement	Multiple incompletely specified or infinite parameter models.	A small number of completely specified models/hypotheses.
Perspective	"All models are wrong, but some are useful."	"Which model is correct?"
Simulation structure	$d \gg n$	$d \ll n$
With increased $n \dots$	Best model grows more complex.	Procedure focuses in on one best model.
Applications		
Context	Exploratory analysis; model selection to address which model will best predict the next sample; imprecise modeling; tapering effects.	Confirmatory analysis; hypothesis testing; model selection to address which model generated the data; Low dimension, precisely specified models.
Ecological examples	Complex model selection applications, e.g., predictive models for community, landscape, and ecosystem ecology; time series applications including forecasting.	Controlled experiments, for instance in physiology/enzymatics/genetics with a limited number of important, well-understood, biological predictors; models including expected or default (null) frameworks, e.g., enzyme kinetics models, Hardy-Weinberg equilibrium, or RAD curves, one of which is expected to be correct.

Notes: The number of parameters in the true model is d ; sample size is n . Abbreviations are: PRESS, predicted residual sum of squares; and RAD, ranked abundance distribution.

Burnham & Anderson 2002, Johnson & Omland 2004, Harrison et al. 2018

A comparison of methods and IC: AIC leads to more complexity than BIC, but predictive ability assessed through cross-validation is similar



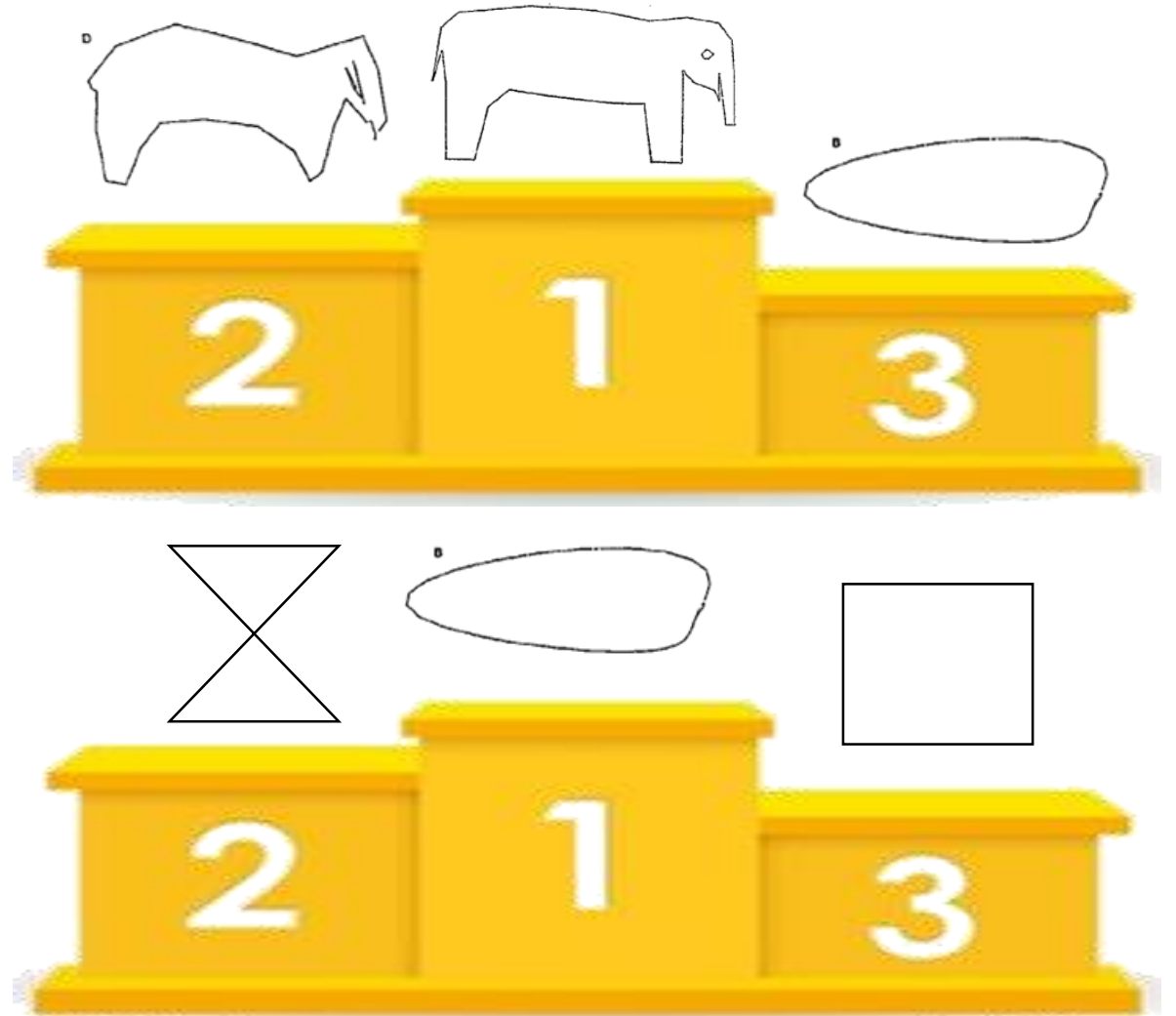
1. stepwise F tests, 2 stepwise AIC, 3 stepwise BIC, 4 best AIC, 5 best BIC, 6 tree regression (1-Se rule), 7 tree regression with minimum risk

Practical recommendations to select the IC

- 👉 Use the **highest-dimensioned model** in the set to make a decision regarding **overdispersion** and **sample size**, etc; and use a single information criterion **consistently**.
- 👉 Make sure you are using the right **number of parameters (K)**. For example, in Ordinary Least Squares (OLS), K includes the intercept and **residual variance**. Some packages do not count the variance as a parameter (`caper::pgls()` vs `phylolm::phylolm()`, <https://github.com/lamho86/phylolm/issues/51>)

About the importance of
**absolute measures of
model quality**

IT will always find a “best” model,
but maybe the “best” model is rubbish



It is crucial to measure **absolute quality** of models, because IT only informs of **relative quality**

- ☞ Assess **goodness of fit**: is the model better than a null model (= intercept only)?
 - Do a likelihood ratio test with the most complex model. IC based on K-L won't select a parsimonious model that doesn't fit (empirical result, not a theorem). Bayesian IC does not have this property (Burnham & Anderson, 2002)
 - Include **null model** in the candidate model set (Mac Nally et al. 2018)
- ☞ **R², model's residuals**, cross-validation
- ☞ A survey found that only 46% of papers using IT provided absolute measures of model quality (Mac Nally et al. 2018)

About the importance of
(not) dredging

What is data dredging?

- ☞ Data dredging: analysing data with few or no a priori questions, by subjectively and iteratively searching the data for patterns and “significance”.
- ☞ Rather than including a large set of variables, develop a sound set of hypotheses, based on the science, *a priori* (=before fitting the models)
- ☞ Data dredging is often regarded as exploratory, rather than confirmatory
- ☞ In any case, be honest and transparent in your papers 😊

About the importance of

**Model averaging
(multimodel inference)**

Model-averaging has received some criticism

- ☞ Cade (2015) criticized the practise of model-averaging coefficients, and the use such coefficients for prediction; on account of different units across models, and collinearity
- ☞ In contrast, averaging the predictions for many models would be OK
(https://atyre2.github.io/2017/06/16/rebutting_cade.html)

About the importance of
**model weights and
“relative importance”**

A reminder: Akaike weights

$$w_i = \frac{\exp(-\frac{1}{2}\Delta_i)}{\sum_{r=1}^R \exp(-\frac{1}{2}\Delta_r)}$$

Probability (weight of evidence) that a model is the actual K-L best model, within the set of candidate models.

Basically, we normalize relative likelihoods to sum up to 1.

AIC	Δ AIC	L(model data)	Akaike weight (w)		
- 2	$-2 - (-2) = 0$	$e^{(-0.5*0)} = 1$	$1 / 1.98 = 0.5$	X1	X1 = 0.81
- 1	$-1 - (-2) = 1$	$e^{(-0.5*1)} = 0.61$	$0.61 / 1.98 = 0.31$	X1 + X2	X2 = 0.31
0	$0 - (-2) = 2$	$e^{(-0.5*2)} = 0.37$	$0.37 / 1.98 = 0.19$	X3	X3 = 0.19

The probability that a given predictor variable appears in the best model is estimated by summing the weights of each model where the variable appears. This has been used as a metric of relative importance.

Limitations of sum of Akaike weights

- ☞ Sum of Akaike weights of predictors are based on a **relative metric of models**, not on a **absolute metric of predictors**, like effect size (coefficient, partitioned R^2)
- ☞ Akaike weights are subject to **sampling variability**: if the data were gathered again, the new sample will most likely generate a different set of weights for the models in the candidate set (Wagenmakers & Farrel 2004)

Simulation is a good tool to
understand the behaviour of
Sum of Akaike weights

➡ 10.000 simulations

➡ $y \sim x_1 + x_2 + x_3 + x_4$

➡ No collinearity

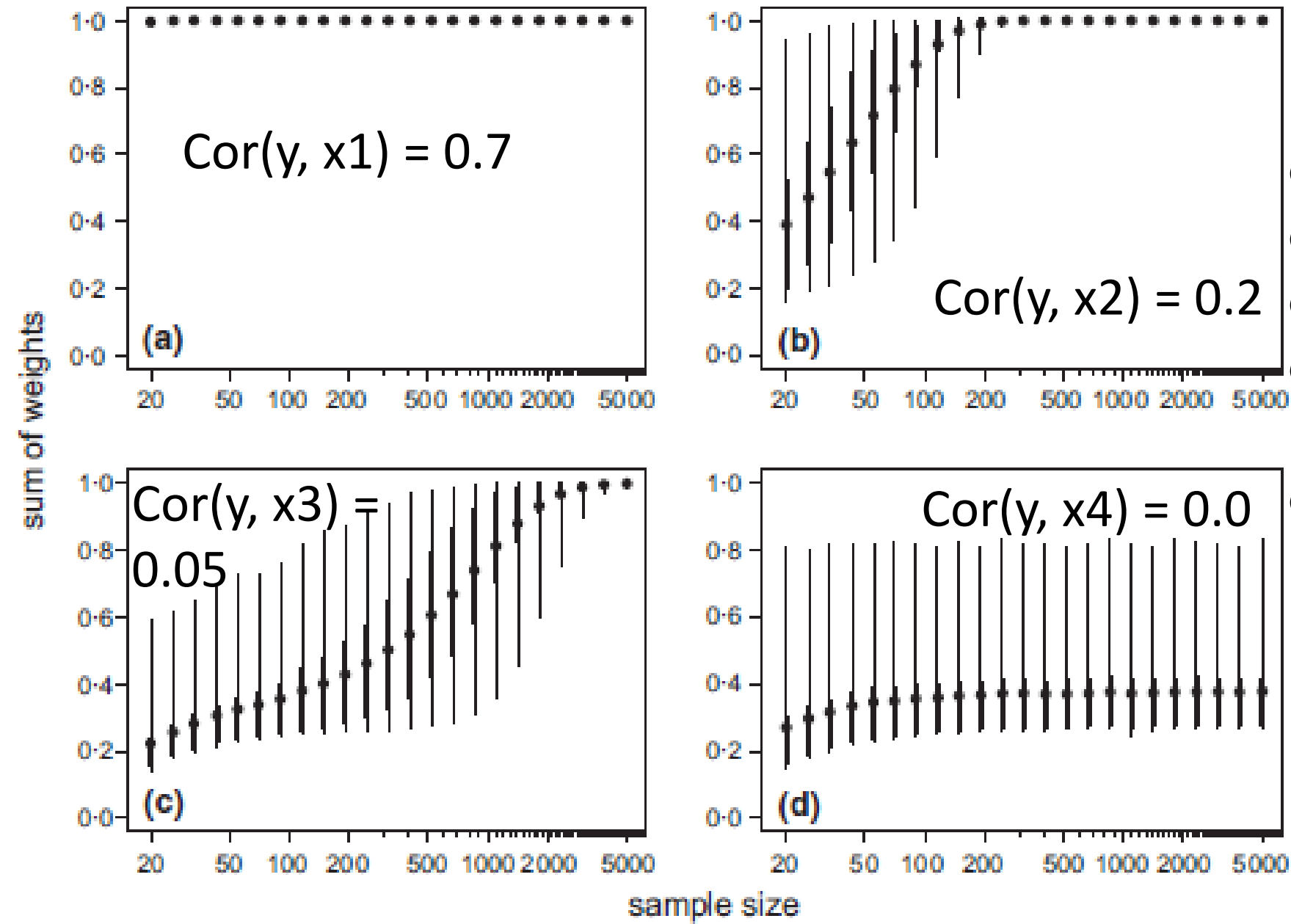
➡ N: 20-5000

➡ $\text{cor}(y, x_1) = 0.7$

➡ $\text{cor}(y, x_2) = 0.2$

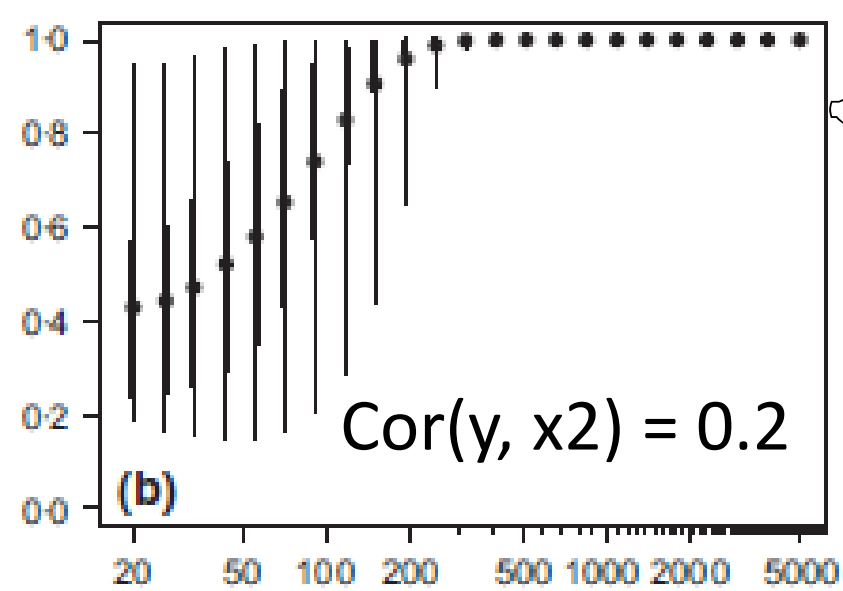
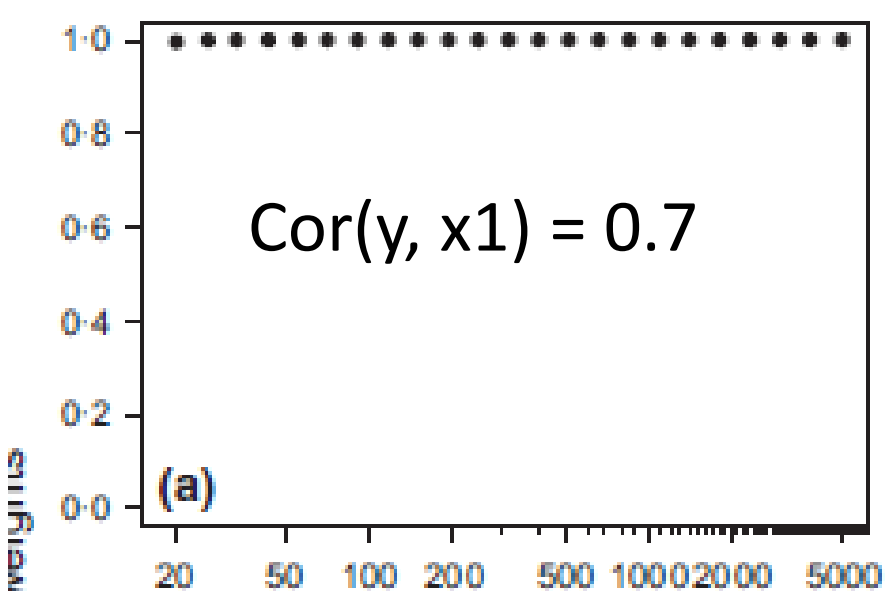
➡ $\text{cor}(y, x_3) = 0.05$

➡ $\text{cor}(y, x_4) = 0.0$

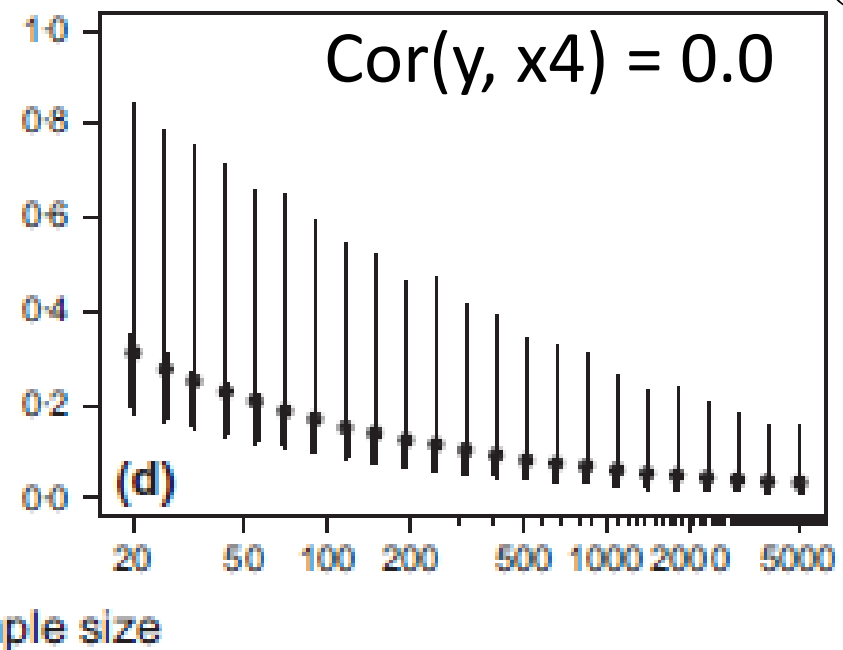
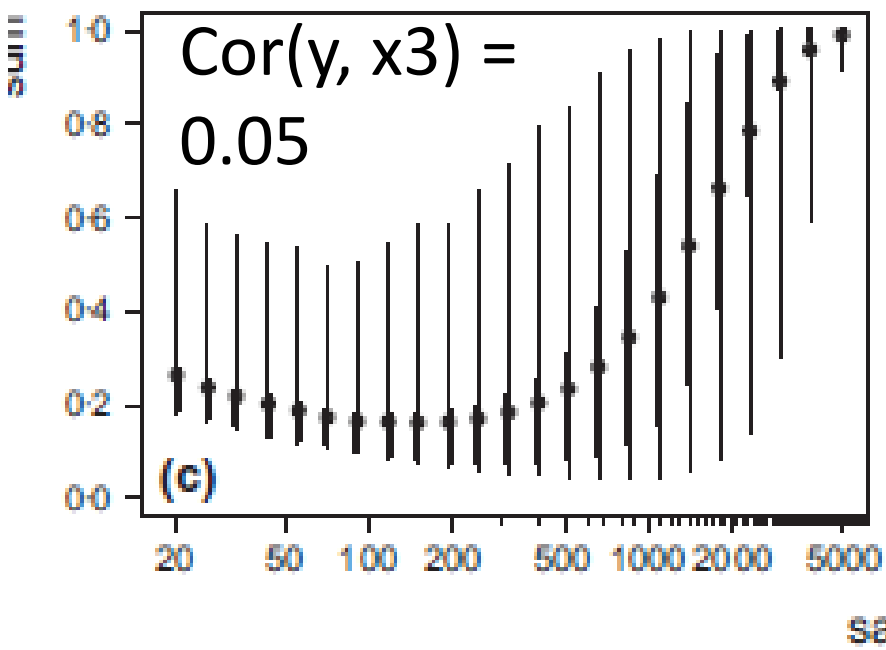


- 10.000 simulations
- $y \sim x1 + x2 + x3 + x4$
- No collinearity
- N: 20-5000

Increasing sample size is not likely to lower the risk of overestimating the support of predictors when using AIC-based sum of weights



✎ BIC penalizes overly complex models. BIC is asymptotically consistent: it selects the correct number of variables.



✎ Under sample sizes commonly found in ecological studies, both AIC and BIC would not provide accurate measure of variable importance

However, what if we focus on
rankings?

- 👉 5 genuine predictors ($x_1 > x_2 > x_3 > x_4 > x_5$), 2 spurious predictors, no collinearity
- 👉 5000 simulations (sample size: 25-1600)

- 5 genuine predictors ($x_1 > x_2 > x_3 > x_4 > x_5$), 2 spurious predictors, no collinearity
- 5000 simulations (sample size: 25-1600)

Modal ranking (the ranking assigned most frequently) was always the correct one (= data generating process)

Increasing sample size improved the ranking of predictors

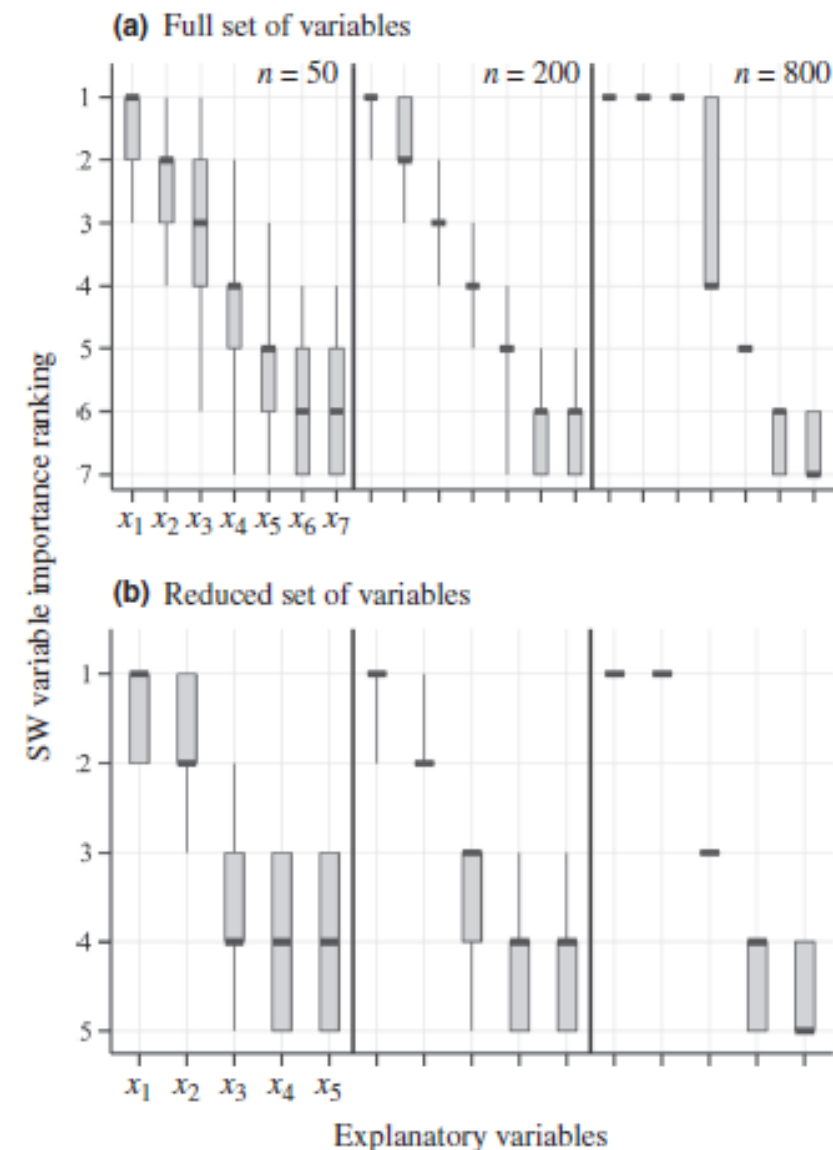
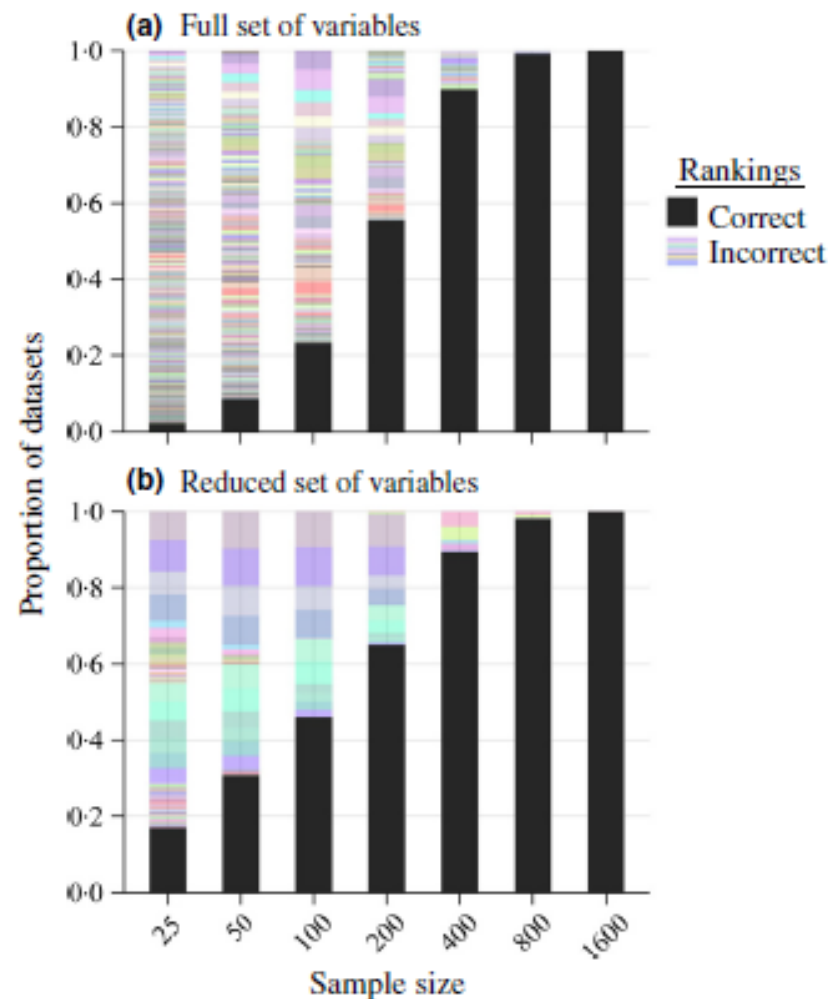


Fig. 1. The proportion of data sets assigned correct and incorrect sets of variable importance rankings using the SW metric. Data sets comprised (a) the full set of 7 candidate explanatory variables and (b) the reduced set of 5 candidate explanatory variables. The black colour bars

What about interactions?

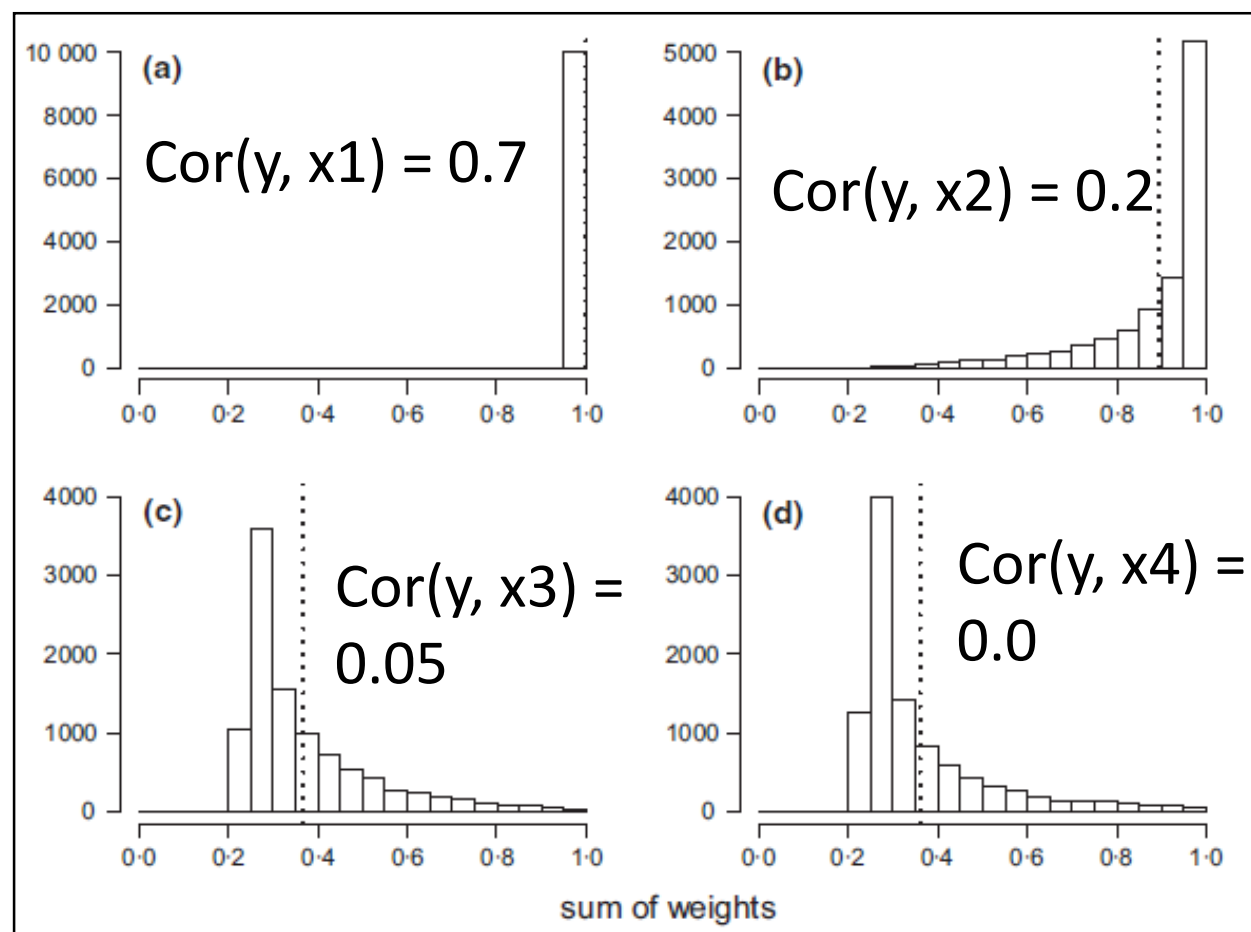
👉 10.000 simulations: $y \sim x_1 + x_2 + x_3 + x_4$

👉 No collinearity

👉 10.000 simulations: $y \sim x_1 + x_2 + x_3 + x_4$

👉 No collinearity

👉 Without interactions



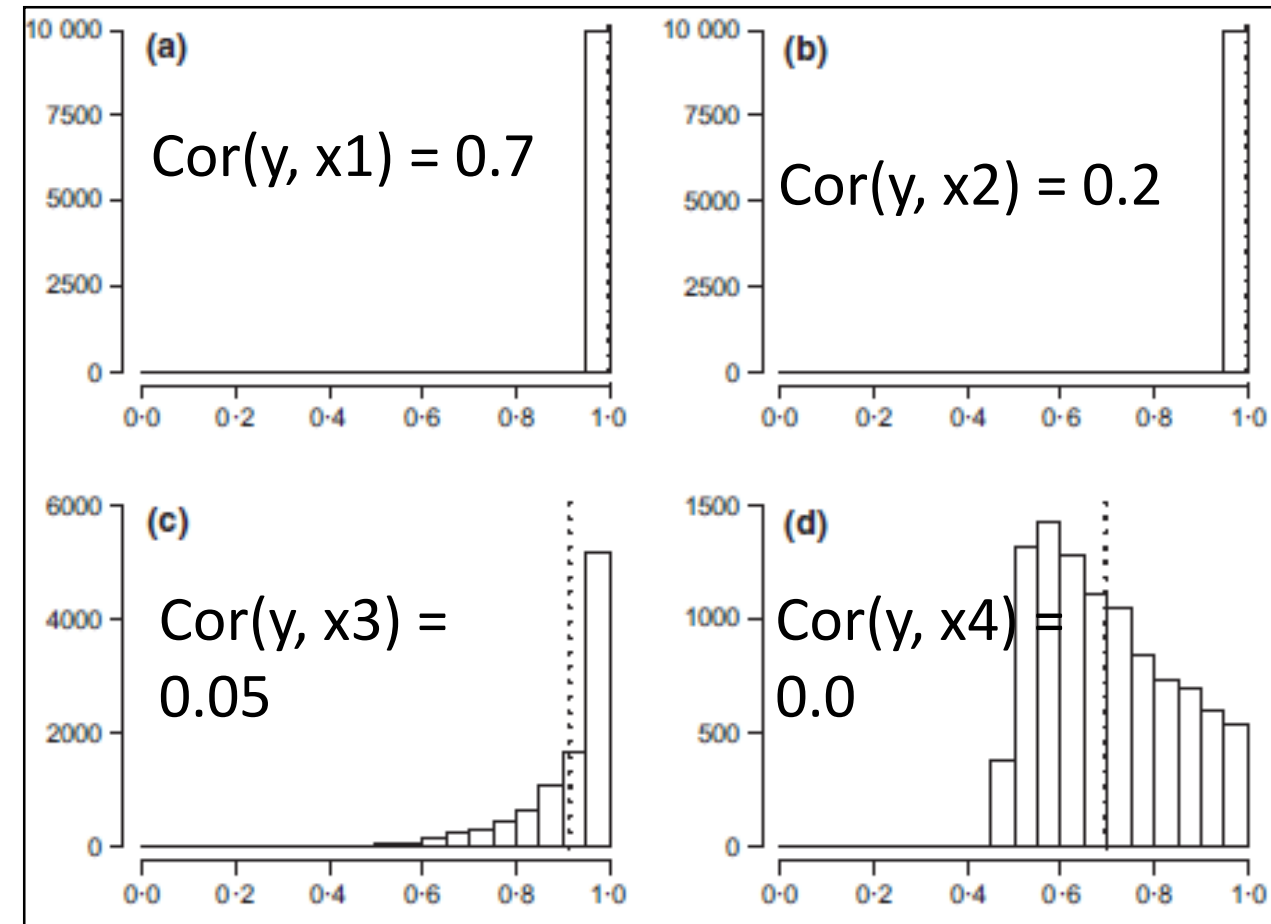
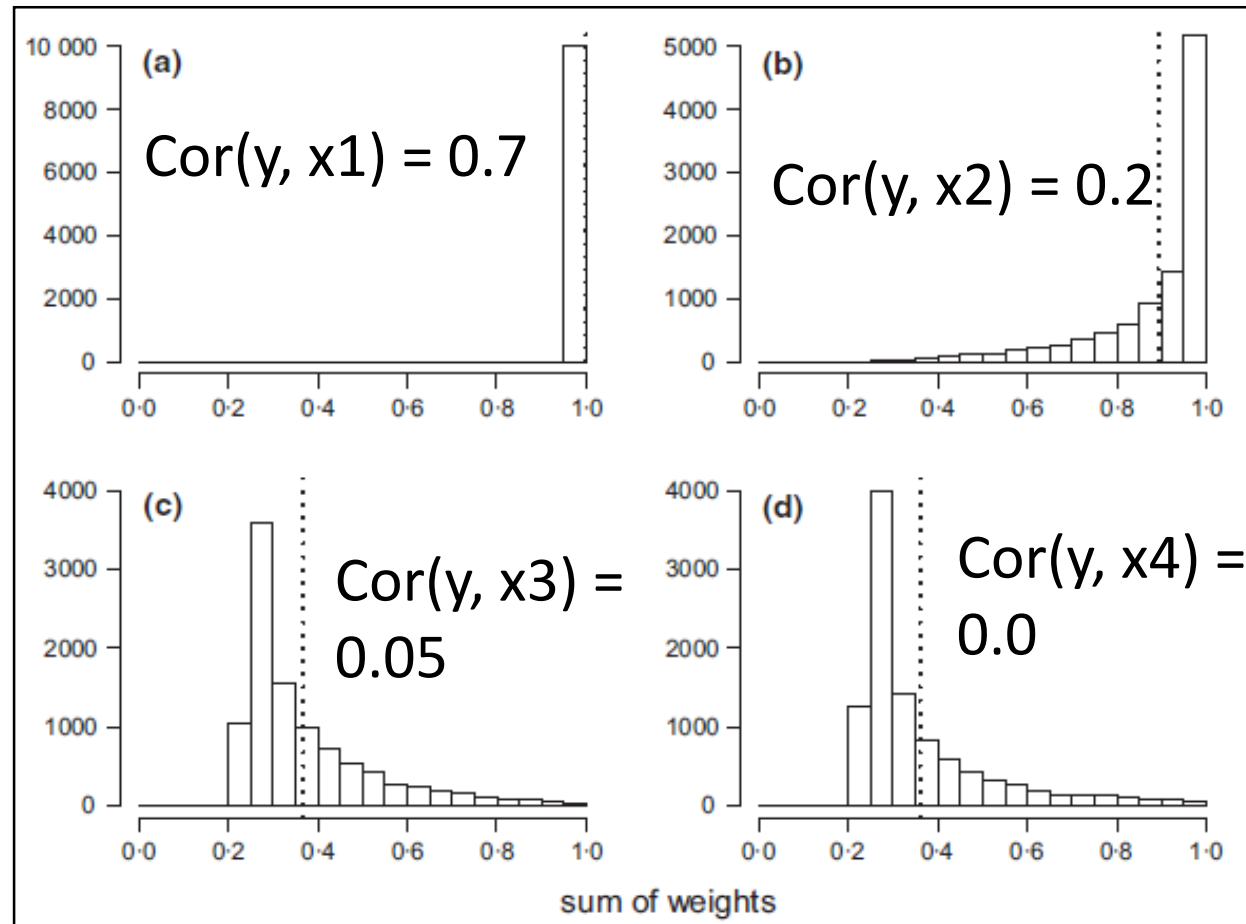
👉 10.000 simulations: $y \sim x_1 + x_2 + x_3 + x_4$

👉 No collinearity

Adding interactions increases the number of models in which the variable appears, increasing SW

👉 Without interactions

👉 Including two-way interactions ($x_1:x_3$, $x_2:x_3$, $x_1:x_2$, $x_3:x_4$)



Simulations also indicate that permutation might not be a solution

- 👉 **Permute** the response variable, or a predictor, and calculate a **Sum of Weights (SW) distribution expected if the predictor had no effect.**
- 👉 Simulation studies with 4 variables (one of them uncorrelated with the response), show that after permuting the response, 95% SW were in the range 0.25-0.82, for all variables. **The lack of a reliable “baseline SW”, which would indicate “no relationship”, makes it difficult to assess variable importance through SW** (Galipaud 2014)
- 👉 **Mean SWs for spurious variables** (variables with an effect size of 0) **are above 0 in regression analyses.** Some studies suggest that the average SW of spurious variables is 0.38 (Galipaud 2017)

About the importance of
**expanding our
options
beyond IT**

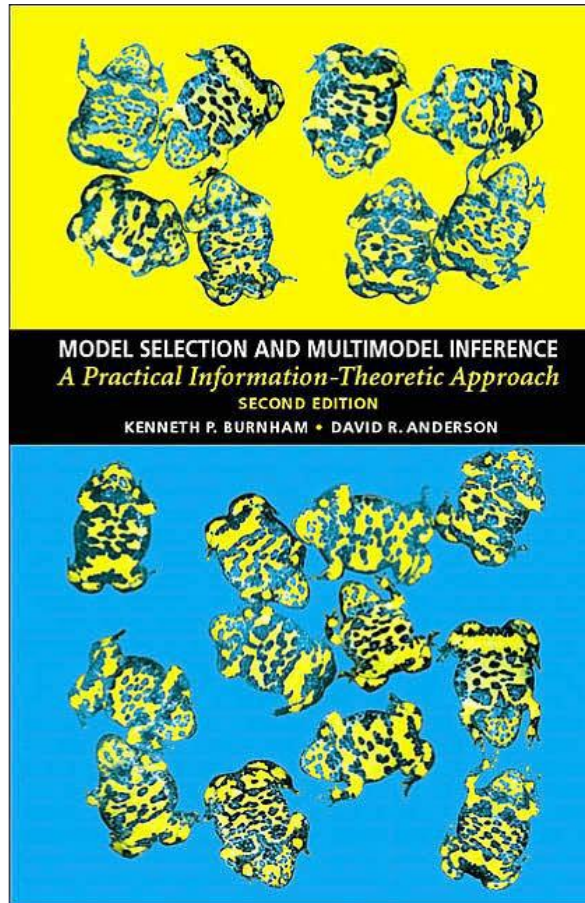


What else is out there?

- 👉 The **bootstrap** and **cross-validation** are alternative model selection methods, although they can be very computer intensive (Burnham & Anderson 2002, Yates et al 2023)
- 👉 **Relative importance** may be assessed through many methods, such as **dominance analysis** (<https://github.com/jluchman/dmir>) or **machine learning** (Pichler & Hartig 2023; Arranz et al. 2025). Note that dominance analysis will not necessarily assign a value of 0 to a variable that is not related to the response.
- 👉 **Shrinkage methods** produce variable estimates in which those that contribute less are shrunk towards 0 (**ridge**, **LASSO**, Dahlgren 2010)
- 👉 There is a **rich paradigm** of **Bayesian methods** and **causally explicit techniques** (Pearl & McKenzie 2018; McElreath 2024)

Suggested reading

Burnham & Anderson 2002



Johnson & Omland 2004



Review

TRENDS in Ecology and Evolution Vol.19 No.2 February 2004

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Model selection in ecology and evolution

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Harrison et al. 2018



A brief introduction to mixed effects modelling and multi-model inference in ecology

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Thank you all!

Any questions?